# GLMCA-Net: Adaptive Feature Enhancement Network Based on Global–Local Multilevel Feature Cascade for Semantic Segmentation of Remote Sensing Images

Changsheng OuYang, Cheng Zhong, and Hui Li

***Abstract*—Remote sensing image semantic segmentation faces challenges such as large intraclass differences, high interclass similarity, complex backgrounds, and significant changes in scale and direction. To address these challenges, this article proposes a novel semantic segmentation network called GLMCA-Net. The encoder of this network adopts a dual-path structure. One path introduces visual state space duality (VSSD) blocks and self-attention mechanisms to achieve long-range dependencies and global perception, while the other path introduces a lightweight residual network to capture fine-grained local information. The features obtained from the two parallel paths are fused through the dual-path complementary fusion (DPCF) block to achieve cross-path feature fusion during the encoding stage. To further enhance the model's perception of small targets and directional changes, we designed an adaptive feature enhancement (AFE) block with an adaptive group-wise rotating convolution (AGRC) module. The AFE block can perform depthwise convolutions at multiple scales and refine the feature maps generated by the visual encoder through receptive fields of different sizes. The AGRC module can dynamically rotate convolution kernels of any size ($K \times K$) to obtain directional information of various targets in remote sensing images. We also propose a lightweight spatial-aware enhanced upsampling (SAEU) block, which combines high-resolution feature maps to restore detailed information and enhance small object recognition capabilities. Extensive experiments on three high-resolution remote sensing datasets demonstrate that GLMCA-Net maintains high-precision segmentation performance while exhibiting excellent computational efficiency and generalization ability. The code will be available at https://github.com/ChangshengOY/GLMCA-Net**

***Index Terms*—Mamba, multilevel feature cascade, remote sensing, semantic segmentation.**

Received 12 June 2025; revised 8 October 2025; accepted 30 October 2025. Date of publication 3 November 2025; date of current version 11 November 2025. This work was supported in part by the Cooperation Projects between the Ministry of Natural Resources and Provincial Governments under Grant 2024ZRBSHZ086, in part by the Key Research and Development Program of Zhejiang under Grant 2024C03236, in part by the Basic Research Program Project of Yunnan Province under Grant 202301AT070173, and in part by Yunnan Institute of Geological Science under Grant KYS-JS2024-005-WT01. *(Corresponding author: Cheng Zhong.)*

Changsheng OuYang and Hui Li are with the School of Earth Sciences, China University of Geosciences, Wuhan 430074, China (e-mail: ouyangcs@cug.edu.cn; rslihui@cug.edu.cn).

Cheng Zhong is with Badong National Observation and Research Station of Geohazards, China University of Geosciences, Wuhan 430074, China (e-mail: zhonglxm@cug.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3628315

## I. Introduction

THE rapid improvement of spatial, spectral, and temporal resolution of remote sensing images has resulted in the expansion of their applications in a variety of fields [1], including environmental protection [2], resource investigation [3], urban planning [4], disaster monitoring [5], and military target identification [6]. This development has created an urgent need for image understanding algorithms with higher accuracy and stronger generalization ability. One of the fundamental tasks in computer vision and remote sensing is the semantic segmentation of remote sensing images. This process involves assigning a precise category label to each pixel in an image, with categories including, but not limited to, buildings, roads, water bodies, and vegetation. This dense, pixel-level categorization enables highly refined analysis and deep understanding of the Earth's surface, the features on it, and their spatial interrelationships.

Despite the significance of semantic segmentation of remote sensing images, its application to high-resolution remote sensing images is encumbered by a series of substantial challenges [7], [8], [9]. As shown in Fig. 1, these images usually exhibit the following characteristics.

*1) Feature Ambiguity:* Images commonly exhibit significant intraclass variation and strong interclass similarity. For instance, buildings of different architectural styles present vastly distinct appearances, while asphalt roads and dark-colored roofs share highly similar textures.

*2) Scale Diversity:* A single scene may simultaneously contain objects with significant scale variations, ranging from large-scale farmland or expansive urban areas to small individual objects such as cars, trees, or building attachments.

*3) Geometric Complexity:* Natural and man-made objects often exhibit irregular boundaries and arbitrary orientations. Traditional fixed convolution kernels struggle to effectively model this geometric variability, often resulting in inadequate feature representation.

*4) Structural Fidelity:* During the decoding and upsampling phases, both large-scale objects and slender objects face risks of structural discontinuity, blurred boundaries, or detail loss. Meanwhile, subtle features of small-scale objects are easily overlooked. Achieving high-fidelity reconstruction of an

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.




Fig. 1. Characteristics exhibited by three well-known remote sensing image datasets. (a) Feature ambiguity. (b) Scale diversity. (c) Geometric complexity. (d) Structural fidelity.

object's overall structure and fine contours remains a critical challenge.

In this article, we will focus on designing an effective network structure to solve the above problems. Deep learning techniques have demonstrated excellent performance in the field of computer vision in recent years due to the exponential growth of learning data and parallel computing resources, as well as continuously optimized deep algorithmic models [10], [11]. These developments have also greatly stimulated the application and expansion of deep learning methods in the field of remote sensing. We also focus our attention here. Despite the noteworthy accomplishments of convolutional neural network (CNN)-based architectures, they are inherently constrained by a significant limitation: the locality of convolution operations imposes boundaries on their receptive fields. This hinders CNNs' capacity to model global contextual information and capture long-range dependencies within images. Despite the introduction of techniques such as atrous convolutions (e.g., DeepLabV3+ [12]) and pyramid pooling modules (PPM) (e.g., PSPNet [13]) to expand the receptive field and aggregate multiscale context, there remains a concern regarding the reliability of contextual information when processing remote sensing images with complex features and significant spectral differences. To address the global context limitations of CNNs, vision transformers (ViTs) [14] and their variants [15], [16] emerged, leveraging self-attention mechanisms to capture global dependencies within images. ViT and its variants have demonstrated exceptional capabilities in modeling long-range dependencies and have become a research hotspot in the field of remote sensing image segmentation. However, processing large-scale, high-resolution remote sensing images presents a considerable challenge because the self-attention mechanism's computational complexity scales quadratically with the number of input image patches (or pixels) [17], [18], [19]. In addition, ViT networks primarily capture global information, making it challenging to preserve fine-grained local details and spatial accuracy [20], [21]. This can compromise both object edge sharpness and structural fidelity.

Recently, Mamba has been proposed and achieved breakthroughs in multiple sequence modeling tasks [22]. It is a novel architecture based on the selective state space model (SSM), characterized by linear computational complexity. By balancing accuracy and efficiency, it emerges as a promising alternative to the transformer. Models such as UNetMamba [23], CDMamba [24], and RSMamba [25] in the visual domain have further expanded their application potential in tasks such as semantic segmentation, change detection, and image classification. UNetMamba introduces the Mamba segmentation decoder (MSD) and local supervision module (LSM) to enhance performance. However, its multiscale processing capabilities heavily rely on the skip connections of the U-Net architecture. RSMamba employs a dynamic multipath activation mechanism to process flattened image sequences, but it primarily focuses on global image understanding and falls short in extracting detailed image information. Most existing studies have focused on validating the feasibility of the Mamba architecture, without fully exploring its potential to simultaneously improve both efficiency and accuracy in remote sensing image analysis. In particular, its application to critical challenges such as multiscale representation, directional variation, and fine-grained detail recovery remains insufficiently investigated.

To address the multifaceted challenges in the field of semantic segmentation of remote sensing images and overcome the limitations of existing methods, this article proposes a novel network called GLMCA-Net. Specifically, GLMCA-Net features a dual-path encoder, comprising a Mamba-transformer branch (MTB) and a residual branch (ResB), to capture global long-range dependencies and extract fine-grained local details, respectively. To effectively integrate these two feature streams, we designed a dual-path complementary fusion (DPCF) block to achieve synergistic interaction between the parallel feature streams. In addition, the decoder incorporates two innovative components. The first is the adaptive feature enhancement (AFE) block, centered around an adaptive group-wise rotating convolution (AGRC) module, to dynamically refine feature discriminability. The second component is the lightweight spatial-aware enhanced upsampling (SAEU) block, designed to ensure the high-fidelity reconstruction of fine-grained objects. The main contributions of this article are summarized as follows.

1) The encoder of GLMCA-Net employs a dual-path configuration, utilizing the MTB and ResB to extract global and local information, respectively. Specifically, MTB integrates the visual state space duality (VSSD) blocks with self-attention mechanisms, effectively capturing global long-range dependencies while reducing computational complexity. Meanwhile, ResB employs residual structures to enhance the expression of fine-grained local details.

2) To fully integrate features extracted from two encoding paths, we propose a novel global–local information fusion module, DPCF block, which incorporates channel attention, spatial attention, and residual connections. The meticulously formulated fusion scheme guarantees complementary interaction between global and local information, thereby attaining a more comprehensive representation of remote sensing image features.

3) The decoder was designed to incorporate two innovative components. The first component is the AFE block, which features the AGRC module at its core. The AFE block enhances feature maps from the visual encoder by performing multiscale depthwise convolutions with varying receptive field sizes. The AGRC module dynamically rotates convolution kernels of any

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

size ($K \times K$) to capture directional information from diverse targets in remote sensing images, thereby enhancing feature discriminability. The second component, the SAEU block, is a lightweight component specifically designed for up-sampling and feature cascading. It integrates high-resolution feature maps from skip connections and incorporates a lightweight yet expressive spatial-aware gate. This design effectively mitigates the contour blurring inherent in traditional upsampling methods, enabling the high-fidelity reconstruction of fine-grained object contours.

## II. Related Work

### A. CNN-Based Semantic Segmentation of Remote Sensing Images

CNNs have evolved in the field of semantic segmentation of remote sensing images, progressing from the early fully convolutional network (FCN) [26] to multiscale encoder–decoder structures. The emergence of FCN revolutionized the field of semantic segmentation, establishing the end-to-end learning paradigm in this domain. Despite its pioneering nature, the continuous downsampling in FCNs led to the loss of spatial detail information and inadequate utilization of contextual information. In order to address this issue, Ronneberger et al. [27] proposed UNet, which introduced skip connections between the encoder and decoder, significantly improving segmentation accuracy. Subsequently, PSPNet [13] introduced the PPM to enhance global context and multiscale information. The DeepLabV3+ [12] further proposed atrous spatial pyramid pooling (ASPP), utilizing parallel atrous convolutions to capture multiscale features. The aforementioned methods have achieved significant progress in natural scene segmentation tasks. However, remote sensing images often contain more complex object categories and more ambiguous target boundaries, which makes it challenging for traditional end-to-end segmentation networks to achieve ideal predictions on remote sensing images.

This series of developments highlights a fundamental contradiction in semantic segmentation: how to achieve a balance between addressing "what" (semantics, typically global) and "where" (location, typically local). Concurrently, as models become increasingly complex in capturing contextual information, their computational complexity often increases accordingly. This evolution also laid the foundation for the subsequent attention received by transformer-based models, owing to their inherent global attention mechanism.

### B. Transformer-Based Semantic Segmentation of Remote Sensing Images

In recent years, the effective utilization of the transformer and its variants in computer vision has gradually expanded to remote sensing image segmentation tasks. Dosovitskiy et al. [14] proposed ViT, which diverges from conventional convolutional modules and utilizes pure self-attention mechanisms to process images. This approach facilitates the direct establishment of global contextual relationships and dependencies, thereby achieving superior performance in classification and segmentation tasks. The emergence of ViT demonstrates that CNNs are not the only way to achieve high-performance visual tasks, opening up a whole new research direction. Its capacity to capture global context is particularly appealing for dense prediction tasks such as semantic segmentation [28]. However, standard ViT exhibits certain limitations when applied directly to semantic segmentation. Specifically, it typically outputs single-scale, low-resolution feature maps, and the computational cost is notably high when processing large-size images. Moreover, ViT's initial success relied on pretraining on large-scale datasets to compensate for its lack of image-specific inductive biases compared to CNNs. To address the high computational cost issue, researchers have focused on designing hierarchical structures [29] or developing efficient attention mechanisms, such as window-based attention [15] and linear attention [30]. However, these methods perform poorly due to their failure to account for the characteristics of remote sensing images, including high intraclass variance and low interclass similarity, complex backgrounds and diverse ground objects, challenges in delineating small targets and elongated structures, and significant variations in scale and orientation [31], [32], [33].

In remote sensing scenarios, previous studies have explored the application of various ViT architectures for semantic segmentation. In their seminal work, Wu et al. [34] proposed Tiff-SegFormer for multisource remote sensing images (RGB + thermal infrared), utilizing a dual encoder to fuse spectral and temperature features. The experimental results demonstrated that Tiff-SegFormer significantly outperformed UNet, DeepLabV3+, HRNet, and single-modal SegFormer. DC-Swin [35] uses the Swin transformer as its backbone to extract contextual information. It employs a decoder composed of densely connected feature aggregation modules to restore features to the original image size, thereby generating remote sensing image segmentation results. Zhu et al. [36] pointed out that for remote sensing images, the CNN methods are constrained by the size of the receptive field. In contrast, ViT-based methods possess global receptive fields but encounter computational bottlenecks when processing long sequences. These efficiency bottlenecks have prompted researchers to explore more efficient attention variants and lightweight designs.

### C. Mamba and Its Visual Variants

Mamba [22] has attracted widespread attention from scholars since its inception. It not only achieves linear complexity but also ensures accuracy through its powerful capabilities for extracting global context information. Recently, Mamba2 [37] further improved the SSM and proposed the concept of state space duality (SSD). It is not merely designed to enhance a specific aspect, such as long-sequence processing. Instead, its goal is holistic efficiency, which includes linear scaling, better inference speed, and the capacity to match or even outperform large-scale transformers with less data and compute power.

Thanks to these features, researchers have developed numerous visual foundation models based on Mamba and Mamba2, such as VMamba [38], DAMamba [39], and VSSD [40]. With the successive release of these models, Mamba-based UNets [41], [42], [43] have gradually achieved remarkable accuracy in medical image segmentation. In the field of remote

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

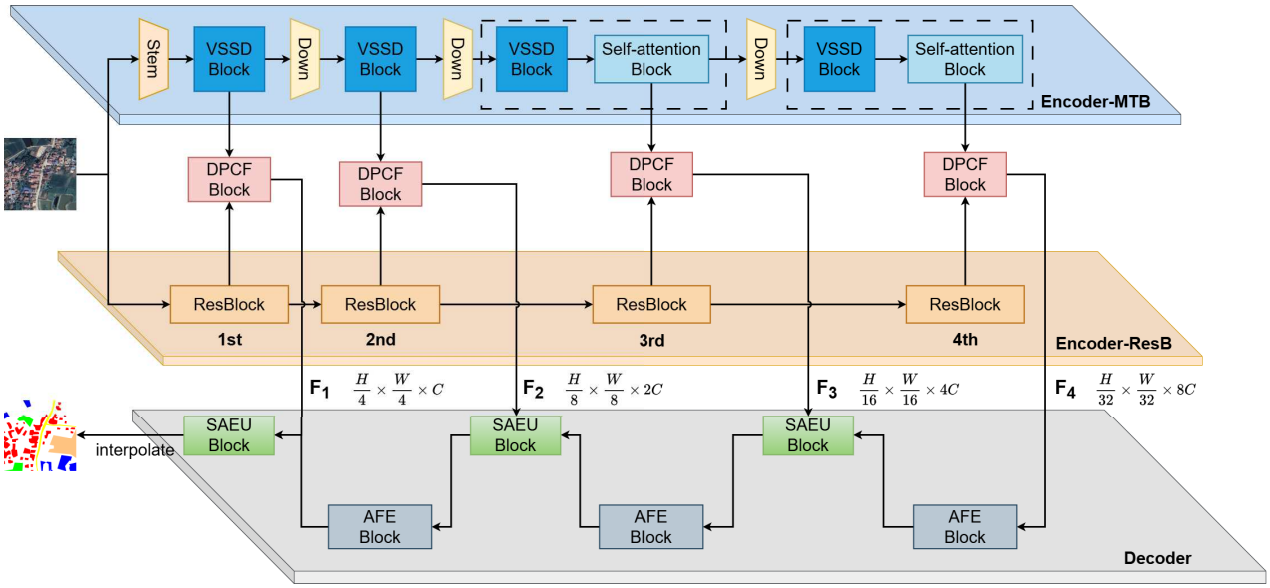



Fig. 2. Overview of the GLMCA-Net.

sensing, an increasing number of researchers are developing new networks based on Mamba models for dense prediction tasks. For example, UNetMamba [23] integrates a Mamba decoder into a CNN encoder to incorporate global information. Samba [36] employs multiple Samba blocks in the encoder to efficiently extract multilevel features. Although previous studies have illustrated the application potential of the Mamba architecture in specific domains, its advantages in terms of synergistic optimization between inference speed and computational accuracy have yet to be systematically explored and validated.

### D. Spiking Neural Networks (SNNs) in Remote Sensing

Another significant frontier in remote sensing research involves exploring brain-inspired computational models, such as SNNs, to achieve ultimate energy efficiency and novel processing capabilities in remote sensing tasks [44]. Recent representative studies have demonstrated the application potential of SNNs in remote sensing from diverse perspectives. Duan et al. [45] focus on the critical needs of on-device computing for platforms like on-orbit satellites and UAVs. They argue that models must not only be highly energy-efficient but also possess online adaptive capabilities to rapidly adjust to environmental changes like weather variations and sensor drift. Their method introduces an efficient unsupervised online adaptation algorithm, adaptive activation scaling (AAS), and a confidence-based instance weighting scheme, achieving leading performance in classification, segmentation, and detection tasks across diverse weather conditions. Niu et al. [46] propose an event-driven SNN with a novel threshold-following reset (TF-reset) and membrane potential modulation for remote sensing image classification. This approach addresses the performance degradation that occurs when converting artificial neural networks (ANNs) to SNNs. Yang et al. [47] propose SeLHIB, a novel self-supervised learning strategy based on high-order information bottleneck theory for SNNs, enabling robust event-based optical flow estimation. This method significantly outperforms traditional ANNs and SNNs in generalization ability and noise robustness while reducing energy consumption by 90.44% and 45.70%, respectively. While the aforementioned works primarily focus on event-driven data or online adaptive scenarios, this article addresses the static semantic segmentation task for remote sensing images. The focus of the article is on global–local multilevel feature modeling and lightweight design. These complementary research directions collectively advance the development of intelligent interpretation of remote sensing data across diverse application scenarios.

## III. Methodology

### A. Model Overview

To address the challenges of high-resolution remote sensing image semantic segmentation, we propose an AFE network based on global–local multilevel feature cascade, termed GLMCA-Net. The overall architecture of GLMCA-Net is illustrated in Fig. 2. Specifically, the remote sensing image $X$ is first processed by two parallel encoders. These encoders are utilized to extract global contextual information and local information, respectively. The corresponding features are then input into the DPCF block for global–local feature fusion, yielding multiscale fused features $F_1, F_2, F_3$, and $F_4$. Subsequently, the feature $F_4$ is input into the AFE block to enhance the directional sensitivity and expressive capability of the feature, yielding the improved feature $F_4^o$. The enhanced feature $F_4^o$ is subsequently input into the SAEU block in conjunction with the shallow feature $F_3$ to achieve feature cascading and upsampling. After this process is repeated three times, the resulting feature undergoes a $1 \times 1$ convolution to yield the final prediction result.

### B. Dual-Path Feature Complementary Encoder

Recognizing that global long-range dependencies and fine-grained local details are complementary for overall image

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

understanding, we designed a novel dual-path encoder architecture. This architecture comprises two parallel-running dedicated backbone networks: MTB and ResB. The MTB employs a hybrid attention strategy that combines the computationally efficient VSSD blocks with standard multihead self-attention (MHSA) blocks. This approach enables the model to capture long-range dependencies and global context information with minimal computational overhead. Meanwhile, the ResB employs a mature CNN, specifically ResNet-18. This dual-path design enables GLMCA-Net to simultaneously perceive broader scene layouts and complex object details, generating rich, multilayered feature representations critical for downstream tasks.

For ResB, it consists of four main stages. Each stage contains multiple residual blocks, which enhance the model's ability to model boundaries and local textures and have good generalization ability and stability. ResB maintains the same spatial scale as the four stages of MTB, which facilitates subsequent fusion and alignment in the decoder.

For MTB, the initial two stages primarily utilize efficient VSSD blocks to process longer sequences and efficiently capture broader context. The final two stages switch to a combination of VSSD blocks and standard MHSA blocks to process feature maps. Since the sequence lengths are significantly reduced at these stages and computational costs are lower, this strategy is more effective for refining features and modeling complex interactions. Furthermore, we innovatively apply learnable scaling factors $\gamma_i$ ($i$ = 1, 2, 3, and 4) to the residual paths in both the VSSD block and the MHSA block. These factors enable the model to dynamically adjust the contribution of each ResB during different training stages, thereby enhancing training stability and feature selection capability.

SSD is a key technology in Mamba2, representing a significant improvement over traditional SSM. It not only enhances the model's expressive power and computational efficiency but also provides a new perspective on understanding the relationship between SSM and the self-attention mechanism. However, the standard SSD mechanism possesses inherent causality, which can impose limitations when dealing with specific types of data, particularly nonsequential visual data. In the process of extracting features from images, the standard SSD block often encounters issues such as limited information acquisition and disrupted spatial continuity.

To solve the above problems, we introduce the VSSD block, as depicted in Fig. 3. The input feature is first processed by a residual block comprising a 3 × 3 depthwise convolution, with the purpose of enhancing the representation of local features [48]. Subsequently, it enters the core noncausal state space duality (NC-SSD) of this module for global context modeling. In the NC-SSD, the role of A undergoes a fundamental shift. Rather than primarily controlling the "forgetting" or "decay" of hidden states, it is repurposed to regulate the degree to which the current information contributes to the global context. This allows the transformation matrix $M$ to be dense, thereby overcoming the challenge of disrupted spatial continuity. In addition, NC-SSD employs a bidirectional scanning strategy to address the issue of limited information acquisition. The mathematical process of NC-SSD can be simply

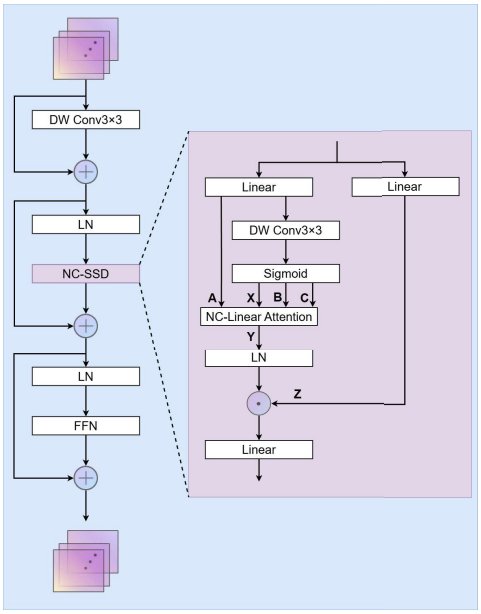



Fig. 3. Detailed structure of the VSSD block.

summarized as follows:

$$H_i = \sum_{j=1}^{i} \frac{1}{A_j} Z_j + \sum_{j=-L}^{-i} \frac{1}{A_{-j}} Z_{-j} = \sum_{j=1}^{L} \frac{1}{A_j} Z_j + \frac{1}{A_i} Z_i, \quad \text{where } Z_j = B_j x(j). \tag{1}$$

$$Y_i = C_i \cdot H_i. \tag{2}$$

Similar to the classic ViT-based models [14], [15], the VSSD block also integrates a feedforward network (FFN) after NC-SSD to improve semantic interaction between channels. The overall process of the VSSD block can be simplified to the following formula:

$$\mathrm{LPU}(X) = X + \mathrm{DWConv}(X) \tag{3}$$

$$X' = \mathrm{LPU}(X) + \mathrm{NC-SSD}(\mathrm{LN}(\mathrm{LPU}(X))) \tag{4}$$

$$\mathrm{VSSD}(X) = X' + \mathrm{FFN}\left(\mathrm{LN}\left(X'\right)\right) \tag{5}$$

where LPU is a 3 × 3 depthwise convolution residual block, and $X$ is the input feature.

### C. DPCF Block

To effectively fuse two distinct types of feature maps from dual-path encoders, we designed the DPCF block to be both lightweight and expressive, ensuring that both global semantics and local details are preserved and mutually enhanced. DPCF block combines channel attention, spatial attention, and residual connections. Channel attention addresses the question of what to fuse. Spatial attention addresses the question of where to fuse. By combining these attention mechanisms with residual connections, the DPCF block ensures that global information is guided to supplement and correct local details, while local details provide concrete evidence to support the global context.

As shown in Fig. 4, the DPCF block initially executes element-wise addition of feature maps from the global and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

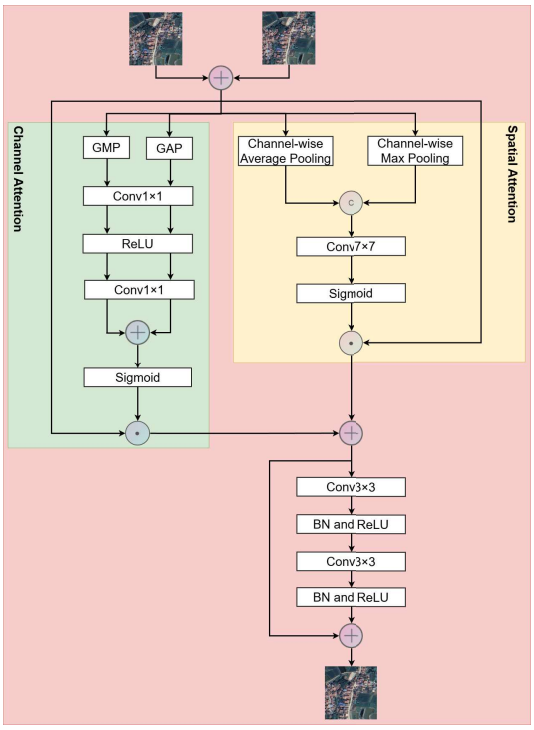



Fig. 4. Detailed structure of the DPCF block.

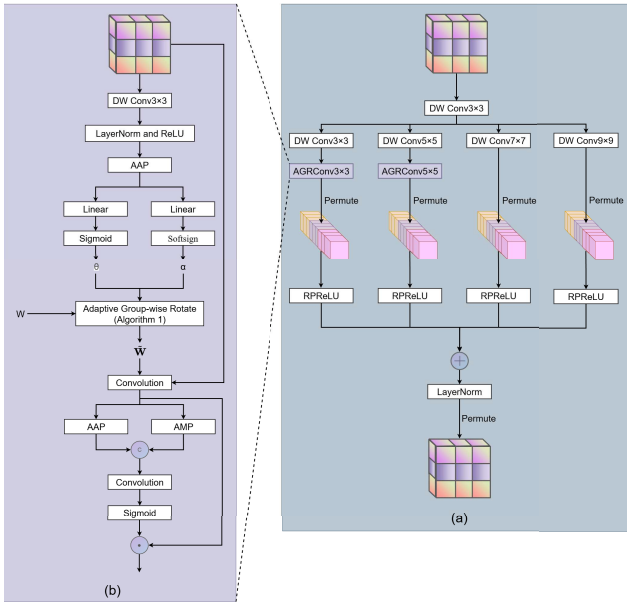



Fig. 5. Detailed structure of the AFE block and AGRC module. (a) AFE block. (b) AGRC module.

local branches, as opposed to concatenation along the channel dimension, with the objective of reducing the channel dimension and computational cost. The fused coarse feature maps are then processed in two parallel attention submodules: the spatial attention branch (SAB) and the channel attention branch (CAB).

The CAB is adapted from the squeeze-excitation (SE) paradigm [49], but we enhance channel attention by capturing more diverse channel information with global average pooling and global max pooling. Specifically, we implement global average pooling and global max pooling on the input features. These two pooling feature maps are then subjected to further enhancement through two shared $1 \times 1$ convolutional layers. Subsequently, these two feature maps are added element-wise and subjected to sigmoid activation to obtain attention weights. Finally, we use the Hadamard product to combine the input features with the attention weights. The CAB emphasizes the most informative channels while suppressing redundant or noisy features. This mechanism is particularly crucial when dealing with objects that exhibit significant variations in texture and appearance, such as buildings or vegetation in remote sensing images. The CAB can be represented by the following equation:

$$\begin{aligned}&\mathrm{CAB}\,(G_i, L_i)\\&= (G_i + L_i) \odot \sigma(\mathrm{Conv}_2(\mathrm{ReLU}(\mathrm{Conv}_1(\mathrm{GMP}(G_i + L_i))))\\&\quad + \mathrm{Conv}_2(\mathrm{ReLU}(\mathrm{Conv}_1(\mathrm{GAP}(G_i + L_i))))) \end{aligned} \tag{6}$$

where $G_i$ and $L_i$ represent global and local features, respectively. The sigmoid activation function is denoted by $\sigma$. Two shared $1 \times 1$ convolution layers are represented by $\mathrm{Conv}_1$ and $\mathrm{Conv}_2$. GMP and GAP denote global max pooling and global average pooling, respectively.

As a complement to the CAB, the SAB focuses on enhancing regions with spatial significance. It generates two 2-D spatial descriptors across the channel dimension. The spatial attention weight is obtained by concatenating these descriptors and passing them through a $7 \times 7$ convolutional layer, which is followed by a sigmoid activation function. The SAB effectively emphasizes regions such as building edges, small cars, or narrow roads, which are often difficult for traditional CNNs to capture due to their irregular and fine-grained nature. The SAB can be represented by the following equation:

$$\begin{aligned}&\mathrm{SAB}\,(G_i, L_i)\\&= (G_i + L_i) \odot \sigma(\mathrm{Conv}_3(C(\mathrm{CAP}\,(G_i + L_i))\\&\quad + \mathrm{CMP}(G_i + L_i)))) \end{aligned} \tag{7}$$

where $\mathrm{Conv}_3$ represents the $7 \times 7$ convolution layer, and CAP and CMP represent the average and maximum values across the channel dimension, respectively.

After dual attention fusion, a small residual convolutional block is applied, comprising two Conv-BN-Activation layers. The final output is a feature map that incorporates both global and local information.

### D. AFE Block

To enhance the model's sensitivity to object orientation and its ability to represent subtle features, we propose the AFE block. Its core concept integrates multiscale deep convolutions with the AGRC module. Specifically, multiscale depthwise convolutions are employed to amplify fine details across different scales, while the AGRC module strengthens geometric modeling and directional sensitivity, thereby ensuring the structural integrity of objects in any orientation.

Fig. 5 depicts the overall structure of the AFE block. The input feature enters the multiscale depthwise convolution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

module after passing through a 3 × 3 depthwise convolution. The introduction of multiscale receptive fields allows the model to capture contextual information from various spatial ranges. The 3 × 3 and 5 × 5 depthwise convolution paths help identify small targets, while the 7 × 7 and 9 × 9 depthwise convolution paths are suitable for perceiving large structures. We have inserted AGRC modules into the 3 × 3 and 5 × 5 paths, which serve as the core enhancement mechanism of this block. Next, the features from the four paths are each passed through an RPReLU activation function to introduce nonlinearity and enhance representational capacity. Finally, the outputs from all branches are summed and fed into a LayerNorm normalization layer, then restored to the original 4-D tensor form as the output.

**Algorithm 1** Adaptive Group-Wise Rotate

**Input:** rotation angles $\theta$, scale factors $\alpha$, and convolution kernel W

**Output:** Dynamic rotating convolution kernel $W'$

1. Generate a canonical grid of coordinates $p_j = \begin{bmatrix} x_j \\ y_j \end{bmatrix}, j = 1, \ldots, N = k^2$
2. Construct a rotation matrix for each group $R_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{bmatrix}, \quad i = 1, \ldots, B \cdot G, R \in R^{B \cdot G \times 2 \times 2}$
3. Apply rotation to all positions $\tilde{p}_{ij} = R_i \cdot p_j + \frac{k-1}{2}$
4. Nearest neighbor and build permutation matrix$idx_{ij} = round(\tilde{p}_{ij}), R_i^{perm}[n, idx_{ij}] = 1$
5. Weight the rotation tensor $R_{rot} = \alpha \odot R^{perm} \in R^{B \cdot G \times N \times N}$
6. Apply rotation to grouped kernel $W' = R_{rot} \cdot W$
7. Reshape back to convolutional format $W' \in R^{B \cdot C_{out} \times C_{in} \times k \times k}$

To achieve rotation perception, the AGRC module first uses a depthwise convolution and global average pooling to compress the spatial information of the input feature $X_{\text{input}} \in R^{B \times C\text{in} \times H \times W}$ into a channel descriptor $X_{\text{pool}} \in R^{B \times C}$. Then, $X_{\text{pool}}$ is processed through a linear layer and Sigmoid normalization to derive the scale factors $\alpha \in R^{B \times G}$ of the rotation convolution kernel. Concurrently, $X_{\text{pool}}$ is processed through another linear layer path to obtain $\theta_{\text{raw}} \in R^{B \times G}$. $\theta_{\text{raw}}$ is then activated by Softsign and multiplied by a fixed angle scaling factor to obtain the final rotation angles $\theta$. The adaptive group-wise rotate algorithm is shown in Algorithm 1. Specifically, we construct a 2-D coordinate grid $P \in R^{N \times 2}$ for the standard convolution kernel and generate a 2-D rotation matrix for each group of $\theta$

$$R_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{bmatrix}, \quad i = 1, \ldots, B \cdot G. \tag{8}$$

Multiply the coordinate vector $P_j$ by the rotation matrix $R_i$, then realign it to the integer coordinate points of the discrete grid using nearest-neighbor discretization. Using the nearest-neighbor discretization, a sparse one-hot permutation matrix $R_i^{\text{perm}}$ is constructed to achieve a new index arranged by angles.

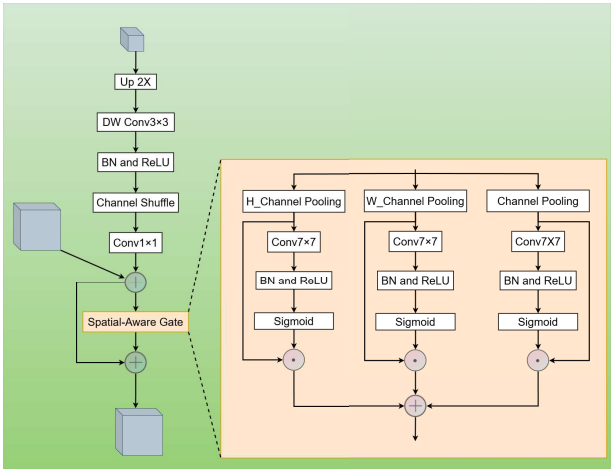



Fig. 6. Detailed structure of the SAEU block.

This process enables the model to dynamically rotate convolutional kernels of arbitrary size ($K \times K$) in a differentiable manner with a clear geometric interpretation. After the rotation matrix is constructed, $\alpha$ is added as scaling factors to control the activation intensity of each rotation kernel. Finally, each group of convolutional kernels is multiplied by the rotation matrix $R_{\text{rot}}$ to obtain the dynamically rotated convolutional kernel $W' \in R^{B \cdot C_{\text{out}} \times C_{\text{in}} \times k \times k}$. To further augment the important regions of different groups, the AGRC module also introduces a lightweight channel attention mechanism when outputting the feature $X_{\text{output}} \in R^{B \times C\text{out} \times H \times W}$.

### *E. SAEU Block*

Traditional upsampling methods, such as transposed convolution or simple bilinear interpolation, often fail to achieve high-fidelity reconstruction. These methods often result in blurred boundaries and the loss of intricate details. To address this issue, we propose a lightweight upsampling and feature cascade block, referred to as the SAEU block. The SAEU module intelligently cascades high-resolution features from the encoder through skip connections. The spatial-aware gate, the core component of the system, is able to recalibrate feature channels in an adaptive manner, thus enabling the network to prioritize the retention of critical boundary cues and structural information during the process of upsampling. This mechanism ensures high-fidelity reconstruction with precise contours of both the overall structure and fine details of the target, as illustrated in Fig. 6.

Specifically, the upsampling process employs a bilinear interpolation technique to restore the size, ensuring the smoothness and continuity of the feature map. Subsequently, depthwise convolution is utilized to enhance the expressiveness of the features while reducing the computational overhead. Finally, the channel shuffle operation is employed to break channel redundancy and promote information exchange between different subchannels. When implementing skip connections, we use pointwise convolution to map the input to the desired output channel dimension, ensuring that the dimensions of the upsampling output align with those of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

skip connection branches. Next, we introduce the spatial-aware gate, which perceives spatially important regions from three dimensions: $H, W$, and the original space. The spatial-aware gate uses a channel fusion strategy combining average pooling and max pooling, along with a $7 \times 7$ convolution layer, to generate an explicit spatial attention map. The spatial attention map is then connected to the input features via a residual connection, achieving adaptive enhancement of the original features. The SAEU block can be simplified to the following equation:

$$F_i' = F_i + \mathrm{Conv}_1(\mathrm{CS}(\mathrm{ReLU}(\mathrm{BN}(\mathrm{DWConv}(\mathrm{UP}\left(F_{i+1}^o\right)))))) \quad (9)$$

$$\mathrm{SAEU}\left(F_i, F_{i+1}^o\right) = F_i' + \mathrm{SAG}(F_i') \quad (10)$$

where $F_i$ is the feature obtained through the DPCF block at stage I, $F_{i+1}^o$ is the feature obtained through the AFE block at stage $i+1$, CS represents the channel shuffle operation, $\mathrm{Conv}_1$ represents pointwise convolution, and SAG represents the spatial-aware gate.

### F. Loss Functions

Due to the widespread class imbalance commonly observed in remote sensing images, conventional single cross-entropy loss often struggles under such conditions. It tends to overfit the dominant class and ignore the minor class targets. Therefore, this article proposes a joint loss function that accounts for category distribution differences. This enhancement of the model's global and local discriminative capabilities is achieved by the proposed function. The joint loss function is defined as follows:

$$L = \lambda_1 L_{\mathrm{ce}} + \lambda_2 L_{\mathrm{dice}} \quad (11)$$

$$L_{\mathrm{ce}} = -\frac{1}{N}\sum\nolimits_{n=1}^{N}\sum_{k=1}^{K} y_k^{(n)} \log \hat{y}_k^{(n)} \quad (12)$$

$$L_{\mathrm{dice}} = 1 - \frac{2}{N}\sum\nolimits_{n=1}^{N}\sum\nolimits_{k=1}^{K} \frac{y_k^{(n)}\hat{y}_k^{(n)}}{y_k^{(n)} + \hat{y}_k^{(n)}} \quad (13)$$

where $N$ represents the total number of pixels in the image, $K$ denotes the total number of categories for classification, $\hat{y}_k^{(n)}$ indicates the model's confidence in predicting that pixel $n$ belongs to category $k$, and $y_k^{(n)}$ represents the true label of the pixel. $L_{\mathrm{ce}}$ denotes the cross-entropy loss, and $L_{\mathrm{dice}}$ denotes the Dice loss. In this article, we set $\lambda_1 = \lambda_2 = 1$, meaning that both losses are equally weighted in the model training process.

## IV. Experimental Settings and Datasets

### A. Datasets

We assess our methodology using three publicly accessible high-resolution remote sensing datasets: the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen dataset, the ISPRS Potsdam dataset, and the LoveDA dataset [50].

*1) ISPRS Potsdam Dataset:* The Potsdam dataset is a widely recognized benchmark dataset that was provided by the ISPRS as part of its 2-D semantic labeling contest. The dataset includes true orthophoto (TOP) images with various spectral combinations and corresponding digital surface models (DSMs). The dataset consists of 38 image tiles of the same size, with a resolution of 0.05 m, each measuring $6000 \times 6000$ pixels. The dataset is manually classified into six common land cover categories: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. We conducted experiments using the RGB three-channel TOP images. Among these, image tiles with IDs 2_11, 2_12, 3_10, 3_11, 3_12, 4_10, 4_11, 4_12, 5_10, 5_11, 5_12, 6_7, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_8, 7_9, 7_11, and 7_12 were utilized as the training set. The image tile with ID 2_10 was used as the validation set. The remaining 14 image tiles were retained as the test set (except for image 7_10, which has incorrect annotations). All image tiles were cropped and padded to patches of size $1024 \times 1024$ pixels.

*2) ISPRS Vaihingen Dataset:* Similar to the Potsdam dataset, the Vaihingen dataset is another benchmark for the ISPRS 2-D semantic labeling contest. It presents a different urban environment from Potsdam, enabling the testing of a model's adaptability to various scenarios. The dataset includes TOP images and DSMs, both with a ground sampling distance (GSD) of 9 cm. The dataset consists of 33 image tiles of varying sizes, encompassing six land cover categories: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. We cropped and padded the near-infrared, R, and G bands of the TOP images into patches of $1024 \times 1024$ pixels for training, validation, and testing.

*3) LoveDA Dataset:* The LoveDA dataset is renowned for its imbalanced category distribution, multiscale objects, and intricate backgrounds. The dataset comprises 5987 optical remote sensing images (including 2653 urban images and 3334 rural images) covering an area of 536.15 $\mathrm{km}^2$, with image sizes of $1024 \times 1024$ pixels and a spatial resolution of 0.3 m. The dataset includes urban and rural scenes from three Chinese cities: Nanjing, Changzhou, and Wuhan. The dataset has been manually annotated with seven land cover categories related to urban and rural environments, including building, road, water, barren, forest, agriculture, and background. The dataset contains 2522 images for training, 1613 images for validation, and 1852 images for testing. Given its coverage of scenarios with pronounced urban-rural disparities, LoveDA has become a crucial benchmark dataset for evaluating the cross-domain generalization capabilities of semantic segmentation models.

### B. Evaluation Metrics

This article uses three mainstream and representative evaluation metrics to comprehensively and objectively assess the performance of the proposed model in remote sensing image semantic segmentation tasks: mean $F1$-score (m$F1$), mean intersection over union (mIoU), and overall accuracy (OA). The following are their calculation formulas:

$$\mathrm{m}F1 = 2 \times \frac{\mathrm{precision} \times \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}} \quad (14)$$

$$\mathrm{precision} = \frac{1}{N}\sum\nolimits_{k=1}^{N} \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k} \quad (15)$$

$$\mathrm{recall} = \frac{1}{N}\sum\nolimits_{k=1}^{N} \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FN}_k} \quad (16)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE RESULTS ON THE ISPRS VAIHINGEN DATASET WITH SOTA METHODS. THE BEST VALUES ARE HIGHLIGHTED IN RED

| Method | Imp. surf. | Building | Low. veg. | Tree | Car | mIoU | mF1 | OA |
|---|---|---|---|---|---|---|---|---|
| PSPNet [13] | 90.03 | 89.91 | 69.27 | 81.67 | 72.31 | 80.64 | 89.03 | 91.12 |
| DeeplabV3+ [12] | 91.80 | 89.55 | 69.09 | 80.12 | 71.27 | 80.37 | 88.82 | 91.74 |
| BANet [54] | 91.99 | 90.48 | 71.41 | 80.56 | 72.10 | 81.31 | 89.43 | 92.07 |
| MANet [51] | 92.25 | 90.02 | 70.54 | 81.66 | 79.91 | 82.88 | 90.44 | 92.28 |
| DC-Swin [35] | 93.12 | 92.87 | 73.61 | 82.76 | 73.86 | 83.24 | 90.61 | 93.23 |
| UNetFormer [55] | 93.05 | 92.23 | 72.09 | 81.89 | 77.91 | 83.43 | 90.75 | 92.93 |
| LANet [52] | 92.34 | 91.48 | 72.26 | 81.79 | 73.53 | 82.28 | 90.04 | 92.61 |
| A2FPN [53] | 93.04 | 92.32 | 71.21 | 81.81 | 78.02 | 83.28 | 90.65 | 92.86 |
| RS3Mamba [56] | 92.81 | 92.34 | 71.95 | 82.29 | 78.11 | 83.50 | 90.79 | 92.93 |
| GLMCA-Net (Ours) | 93.80 | 93.43 | 74.07 | 83.13 | 84.57 | 85.80 | 92.19 | 93.62 |

$$\text{mIoU} = \frac{1}{N}\sum_{k=1}^{N}\frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k + \text{FN}_k} \tag{17}$$

$$\text{OA} = \frac{\sum_{k=1}^{N}\text{TP}_k}{\sum_{k=1}^{N}(\text{TP}_k + \text{FP}_k + \text{TN}_k + \text{FN}_k)} \tag{18}$$

where $N$ denotes the number of classes. $\text{TP}_k$ represents the number of pixels correctly predicted by the model as belonging to class $k$, $\text{FP}_k$ represents the number of pixels incorrectly predicted by the model as belonging to class $k$, $\text{TN}_k$ represents the number of pixels correctly predicted by the model as not belonging to class $k$, and $\text{FN}_k$ represents the number of pixels incorrectly predicted by the model as not belonging to class $k$.

### C. Implementation Details

All experiments in this study were performed on a cloud computing platform configured with an NVIDIA GeForce RTX 3090 GPU (24-GB VRAM) and an Intel Xeon Platinum 8362 CPU operating at 2.80 GHz. The experimental environment was implemented using the PyTorch 2.5.0 deep learning framework and CUDA 12.4 as the GPU acceleration backend, ensuring good computational efficiency and compatibility. The model training utilized the AdamW optimizer and cosine strategy to adjust the learning rate, with a base learning rate set to $6e^{-4}$ and a smaller learning rate of $6e^{-5}$ for the backbone network. During the training phase, the batch size was set to 8, and the number of training epochs was set to 105. The input image resolution was uniformly set to $1024 \times 1024$. Various data augmentation techniques were employed to enhance the model's robustness in handling small objects and complex backgrounds. These techniques included random cropping, flipping, color jittering, and mosaic augmentation.

## V. EXPERIMENTAL RESULTS AND ANALYSIS

### A. Results on the ISPRS Vaihingen Dataset

We evaluate the effectiveness and performance of GLMCA-Net using the Vaihingen dataset. The selected comparison methods include convolution-based methods such as PSPNet [13], DeeplabV3+ [12], MANet [51], LANet [52], and A2FPN [53], transformer-based methods such as BANet [54], DC-Swin [35], and UNetFormer [55], and Mamba-based methods such as RS3Mamba [56]. Table I presents the quantitative evaluation results of various state-of-the-art (SOTA) methods on the Vaihingen dataset, and our proposed GLMCA-Net achieves the best performance across all metrics. Specifically, our GLMCA-Net achieves improvements of 2.37% and 2.30% in mIoU, 1.44% and 1.40% in m$F1$, and 0.69% and 0.69% in OA, respectively, when compared with recent methods based on the transformer-based methods (e.g., UNetFormer) and the Mamba-based methods (e.g., RS3Mamba). From the segmentation results of various objects, GLMCA-Net is particularly outstanding in the identification of small targets such as cars. In the car category, GLMCA-Net exhibits an mIoU of 84.57%, which surpasses the 80.01% mIoU of MANet by 4.66% and is significantly ahead of other methods. This clearly demonstrates that the AFE block we designed plays a crucial role in enhancing the model's perception and fine segmentation capabilities for small objects. In addition, in large-area regions such as impervious surfaces, buildings, and low vegetation, GLMCA-Net also exhibits higher category consistency and boundary recognition capability. This observation validates the efficacy of the dual-path encoding structure and multilevel feature fusion strategy.

Fig. 7 shows a visual comparison of the segmentation outputs produced by GLMCA-Net and other methods on the Vaihingen dataset. Specifically, for the first image, due to the presence of shadows and the diversity of car body colors, all comparison models exhibit varying degrees of boundary ambiguity for cars within the two red-boxed areas. In contrast, GLMCA-Net accurately identifies the car with clear boundaries. For the second image, GLMCA-Net's capacity to discern fine-grained local information and global long-range dependencies results in a more complete extraction of building shapes and clearer boundaries. Even the low vegetation under the building's shadow is accurately identified. For the third image, GLMCA-Net accurately identifies the complete background. However, models such as UNetFormer and RS3Mamba not only misclassify the background as impervious surfaces but also incorrectly identify part of the road as buildings. Consequently, many models tend to produce misclassifications when the image contains a large number of densely distributed and spectrally similar cluttered objects. In contrast, GLMCA-Net is capable of effectively alleviating this issue to a certain extent. The aforementioned results provide substantial validation of the rationality of GLMCA-Net's structural design, which effectively enhances interclass discrimination capability and intraclass consistency. This design demonstrates stronger robustness and generalization ability in complex scenarios.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

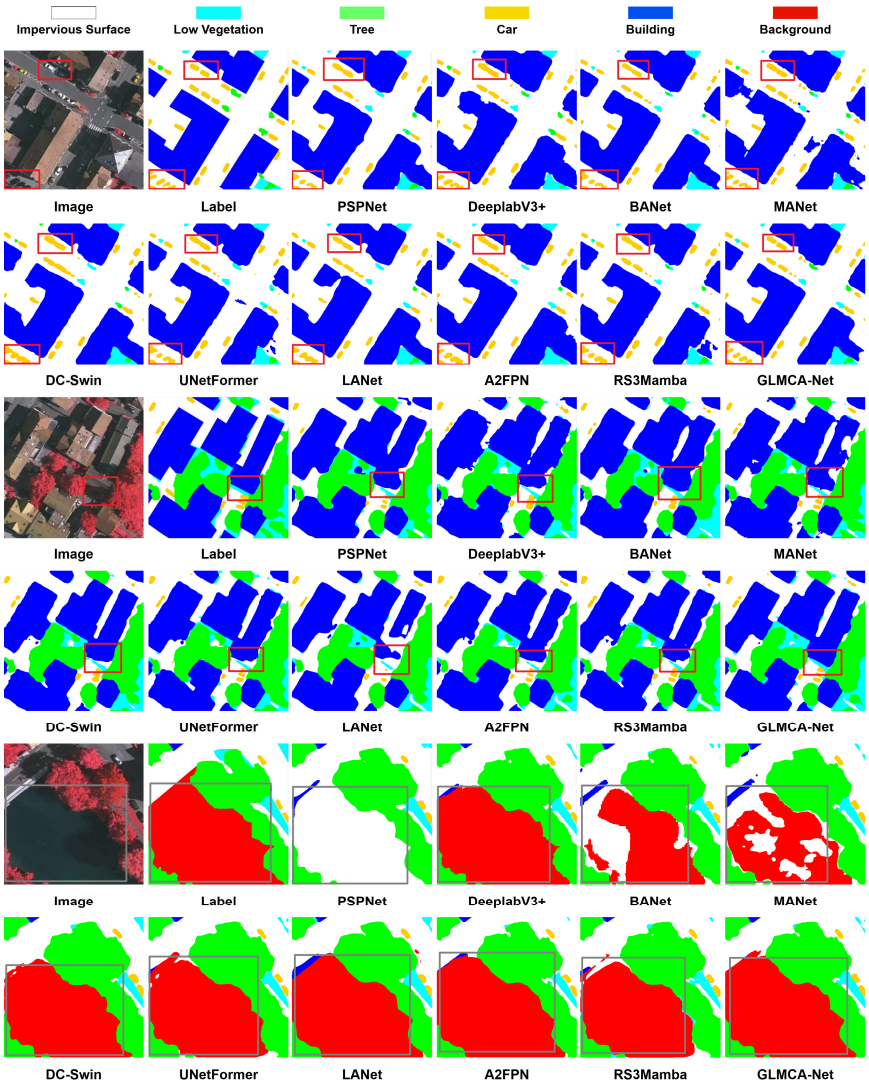



Fig. 7. Visual comparison of the segmentation outputs for the ISPRS Vaihingen dataset.

TABLE II
QUANTITATIVE RESULTS ON THE ISPRS POTSDAM DATASET WITH SOTA METHODS. THE BEST VALUES ARE HIGHLIGHTED IN RED

| Method | Imp. surf. | Building | Low. veg. | Tree | Car | mIoU | mF1 | OA |
|---|---|---|---|---|---|---|---|---|
| PSPNet [13] | 82.41 | 89.27 | 73.40 | 74.14 | 89.96 | 81.83 | 89.84 | 88.12 |
| DeeplabV3+ [12] | 85.29 | 89.80 | 73.65 | 76.00 | 97.28 | 82.40 | 90.21 | 89.18 |
| BANet [54] | 87.42 | 91.29 | 74.07 | 77.17 | 90.18 | 84.03 | 91.16 | 90.01 |
| MANet [51] | 86.26 | 91.01 | 75.83 | 78.56 | 91.99 | 84.73 | 91.60 | 90.11 |
| DC-Swin [35] | 88.10 | 92.70 | 76.86 | 78.09 | 89.41 | 85.03 | 91.78 | 90.97 |
| UNetFormer [55] | 86.76 | 91.38 | 75.99 | 78.62 | 88.84 | 84.32 | 91.37 | 90.42 |
| LANet [52] | 87.30 | 92.06 | 76.24 | 78.03 | 90.97 | 84.92 | 91.71 | 90.53 |
| A2FPN [53] | 86.89 | 91.32 | 76.12 | 78.55 | 90.61 | 84.70 | 91.59 | 90.39 |
| RS3Mamba [56] | 86.88 | 92.47 | 74.82 | 77.90 | 92.34 | 84.88 | 91.65 | 90.26 |
| GLMCA-Net (Ours) | 88.61 | 93.63 | 77.12 | 79.60 | 92.92 | 86.37 | 92.54 | 91.35 |

### *B. Results on the ISPRS Potsdam Dataset*

In order to further validate the effectiveness and performance of GLMCA-Net, experiments were conducted on the Potsdam dataset. Table II shows the quantitative evaluation results of our method compared with various SOTA methods on the Potsdam dataset. As can be observed, GLMCA-Net achieved the best performance, significantly outperforming current mainstream semantic segmentation methods. Specifically, GLMCA-Net achieved an mIoU of 86.37%, which is 1.34% and 1.49% higher than the latest segmentation models, DC-Swin and RS3Mamba, respectively. It also achieved an m$F1$ of 92.54%, yielding improvements of 0.76% over DC-Swin and 0.89% over RS3Mamba. In terms of OA, GLMCA-Net achieved 91.35%, representing an increase of 1.09% compared to RS3Mamba, demonstrating the advantage of our method in overall classification accuracy. With regard

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

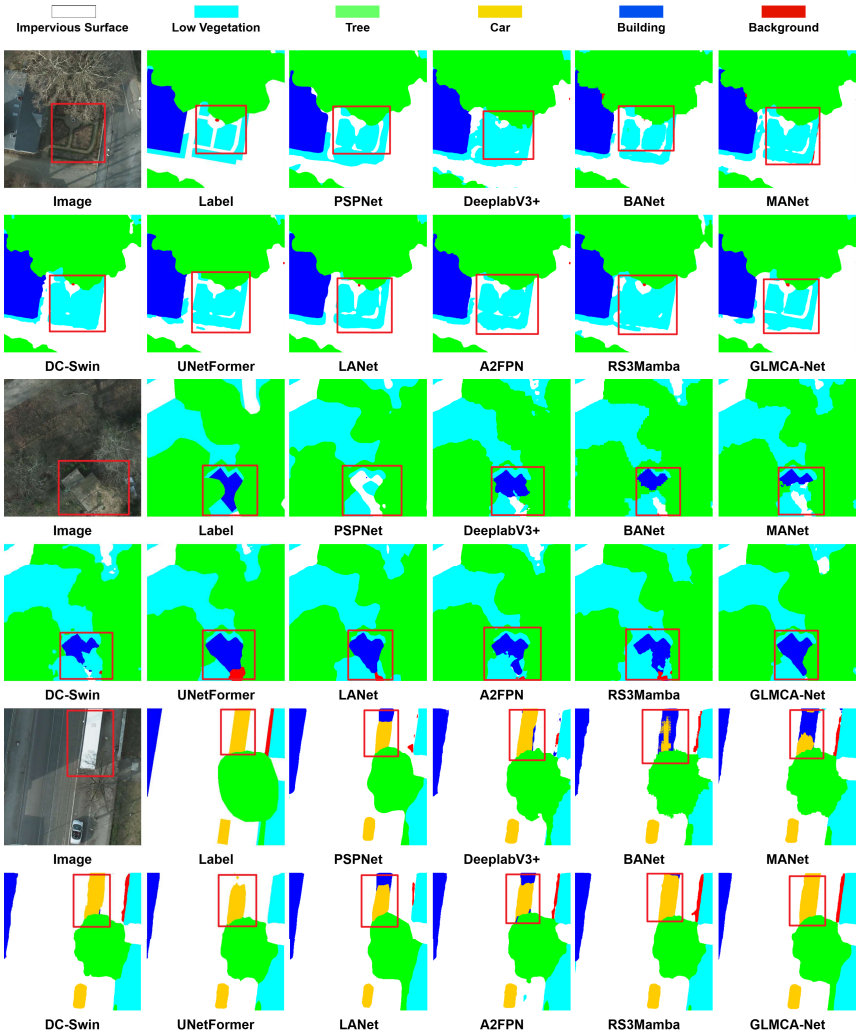



Fig. 8. Visual comparison of segmentation outputs for the ISPRS Potsdam dataset.

to performance across categories, GLMCA-Net demonstrates particularly strong results on two large-area and highly heterogeneous categories: buildings (93.63%) and impervious surfaces (88.61%). It also achieves excellent performance on the small-object category of cars, reaching an mIoU of 92.92%. These results are highly consistent with the performance observed on the Vaihingen dataset, further validating the effectiveness of the proposed global–local multilevel feature cascade structure in modeling global spatial relationships and extracting fine-grained boundaries. They also highlight the significant role of the AFE block and SAEU block in enhancing the model's capability for small-object perception and precise localization.

Fig. 8 shows a visual comparison of the segmentation outputs produced by GLMCA-Net and other methods on the Potsdam dataset. Specifically, for the first image, models such as PSPNet, DeeplabV3+, MANet, UNetFormer, and A2FPN exhibited a certain degree of misclassification when distinguishing between low vegetation and impervious surfaces. In contrast, GLMCA-Net demonstrates greater accuracy in extracting low vegetation boundaries, showcasing stronger category discrimination and boundary perception capabilities. For the second image, the presence of branches obstructing the view above the buildings results in the misclassification of the building regions by all comparison models. However, GLMCA-Net accurately identifies building shapes with clear boundaries and intact structures, demonstrating its robustness and structural perception capabilities in complex scenes. For the third image, GLMCA-Net accurately identifies the extra-long truck, while networks like LANet and MANet mistakenly classify it as a building. These results indicate that, owing to the effectiveness of its structural design, GLMCA-Net is capable of achieving more accurate object recognition and boundary localization in complex scenes. This significantly enhances the overall quality of the segmentation results.

### C. Results on the LoveDA Dataset

The LoveDA dataset presents significant challenges due to its pronounced urban-rural disparities, unequal category distribution, and frequent occlusions and lighting changes. These complex scenarios impose higher demands on the model's category discrimination capability, boundary perception capability, and cross-domain robustness. Table III presents the quantitative evaluation results of our proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE RESULTS ON THE LOVEDA DATASET WITH SOTA METHODS. THE BEST VALUES ARE HIGHLIGHTED IN RED

| Method | Background | Building | Road | Water | Barren | Forest | Agriculture | mIoU |
|---|---|---|---|---|---|---|---|---|
| PSPNet [13] | 44.39 | 54.99 | 46.56 | 75.38 | 20.57 | 45.00 | 63.84 | 50.10 |
| DeeplabV3+ [12] | 40.65 | 53.96 | 51.74 | 75.54 | 18.69 | 44.85 | 62.33 | 49.68 |
| BANet [54] | 43.56 | 55.18 | 51.50 | 78.41 | 15.64 | 45.34 | 59.55 | 49.88 |
| MANet [51] | 45.53 | 55.29 | 55.01 | 78.91 | 17.76 | 45.15 | 64.71 | 51.76 |
| DC-Swin [35] | 43.77 | 58.26 | 57.44 | 80.17 | 20.06 | 46.33 | 58.34 | 52.05 |
| UNetFormer [55] | 45.61 | 57.52 | 56.77 | 79.79 | 18.64 | 47.52 | 62.43 | 52.61 |
| LANet [52] | 42.58 | 58.44 | 56.73 | 80.47 | 19.53 | 45.32 | 57.54 | 51.52 |
| A2FPN [53] | 45.03 | 56.61 | 53.18 | 78.79 | 19.22 | 45.51 | 62.19 | 51.50 |
| RS3Mamba [56] | 40.44 | 55.75 | 50.06 | 78.50 | 17.15 | 46.20 | 64.09 | 50.31 |
| GLMCA-Net (Ours) | 45.71 | 59.46 | 58.55 | 81.10 | 22.40 | 46.99 | 63.13 | 53.91 |

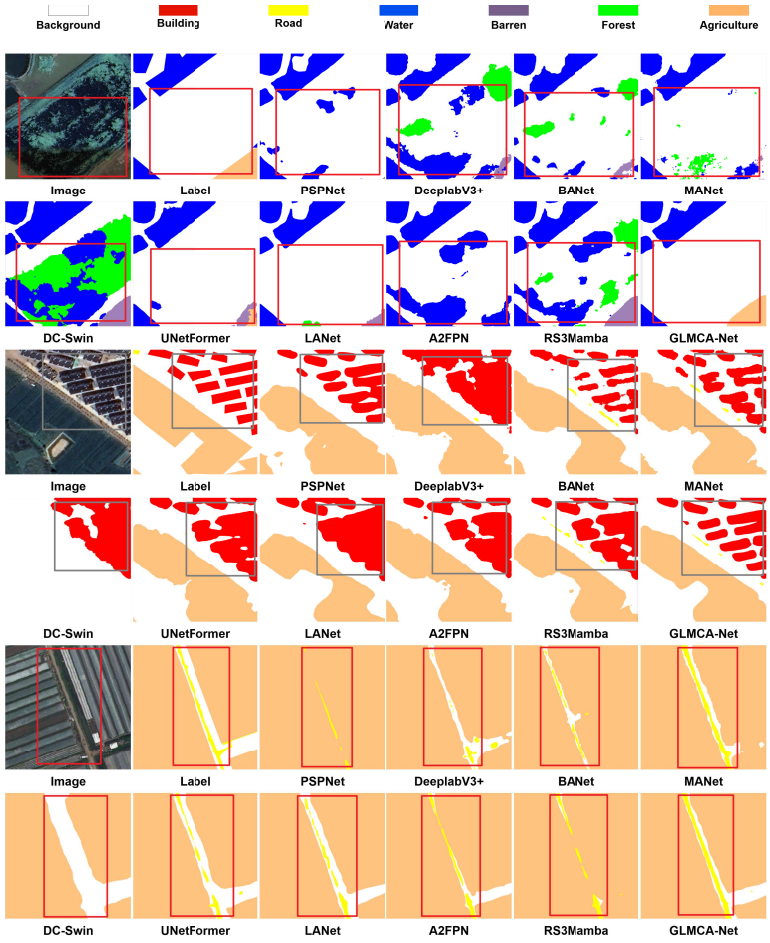



Fig. 9. Visual comparison of segmentation outputs for the LoveDA dataset.

GLMCA-Net compared with various SOTA methods on the LoveDA dataset. GLMCA-Net achieved the highest score of 53.91% on the mIoU metric, demonstrating its superior global modeling and local detail extraction capabilities in complex remote sensing scenarios. In terms of specific category performance, GLMCA-Net achieved leading performance in key categories such as buildings (59.46%), roads (58.55%), water (81.10%), and barrens (22.40%). It is noteworthy that the barren category accounts for an extremely low proportion of the training set. Due to its irregular terrain and complex texture, it is easily confused with other categories, posing a significant challenge to the models. In this highly imbalanced category context, GLMCA-Net outperformed UNetFormer by 3.76% and RS3Mamba by 5.25%. Our method demonstrates particularly outstanding performance. This indicates that GLMCA-Net's dual-path encoding structure can accurately capture both global dependencies and local detail features, while the SAEU block effectively restores spatial details. The structural design of GLMCA-Net affords it superior generalizability when confronted with imbalanced category scenarios.

Fig. 9 shows a visual comparison of the segmentation outputs produced by GLMCA-Net and other methods on the LoveDA dataset. Specifically, for the first image, all

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION STUDY RESULTS ON THE ISPRS VAIHINGEN DATASET

| Method | MTB | ResB | DPCF block | AFE block | SAEU block | Imp. surf. | Building | Low. veg. | Tree | Car | mIoU (%) | mF1 (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | √ | | | | | 87.94 | 81.39 | 60.58 | 74.60 | 39.94 | 68.89 | 80.26 |
| B | | √ | | | | 91.26 | 89.33 | 69.95 | 80.99 | 70.54 | 80.41 | 88.86 |
| C | √ | √ | | | | 92.36 | 90.54 | 70.61 | 81.38 | 75.34 | 82.04 | 89.90 |
| D | √ | √ | √ | | | 92.38 | 91.30 | 71.28 | 81.91 | 78.55 | 83.08 | 90.55 |
| E | √ | √ | √ | √ | | 93.23 | 92.31 | 72.37 | 82.24 | 81.30 | 84.29 | 91.28 |
| F | √ | √ | √ | √ | √ | 93.80 | 93.43 | 74.07 | 83.13 | 84.57 | 85.80 | 92.19 |

TABLE V
ABLATION STUDY RESULTS ON THE ISPRS POTSDAM DATASET

| Method | MTB | ResB | DPCF block | AFE block | SAEU block | Imp. surf. | Building | Low. veg. | Tree | Car | mIoU (%) | mF1 (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | √ | | | | | 79.21 | 82.68 | 64.90 | 60.25 | 82.86 | 73.98 | 84.69 |
| B | | √ | | | | 83.97 | 87.90 | 72.51 | 75.33 | 90.31 | 82.01 | 89.95 |
| C | √ | √ | | | | 85.16 | 90.01 | 74.35 | 76.45 | 90.98 | 83.39 | 90.79 |
| D | √ | √ | √ | | | 86.30 | 90.41 | 76.09 | 77.28 | 91.29 | 84.27 | 91.33 |
| E | √ | √ | √ | √ | | 88.05 | 92.36 | 76.06 | 79.03 | 92.55 | 85.61 | 92.10 |
| F | √ | √ | √ | √ | √ | 88.61 | 93.63 | 77.12 | 79.60 | 92.92 | 86.37 | 92.54 |

TABLE VI
ABLATION STUDY RESULTS ON THE LOVEDA DATASET

| Method | MTB | ResB | DPCF block | AFE block | SAEU block | Background | Building | Road | Water | Barren | Forest | Agriculture | mIoU (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | √ | | | | | 37.77 | 46.75 | 43.01 | 71.33 | 13.32 | 41.66 | 57.33 | 44.45 |
| B | | √ | | | | 43.61 | 54.42 | 53.33 | 77.52 | 18.23 | 47.32 | 60.82 | 50.75 |
| C | √ | √ | | | | 43.35 | 55.25 | 54.00 | 78.62 | 19.64 | 46.65 | 61.76 | 51.32 |
| D | √ | √ | √ | | | 43.05 | 57.93 | 53,83 | 80.34 | 18.12 | 48.74 | 63.81 | 52.26 |
| E | √ | √ | √ | √ | | 45.36 | 57.97 | 57.80 | 80.91 | 18.55 | 47.44 | 63.63 | 53.10 |
| F | √ | √ | √ | √ | √ | 45.71 | 59.46 | 58.55 | 81.10 | 22.40 | 46.99 | 63.13 | 53.91 |

comparison models incorrectly classified the background category as water and failed to identify the agriculture category in the bottom-right corner. In contrast, GLMCA-Net not only effectively avoided these misclassifications but also accurately identified the agriculture category, demonstrating its exceptional semantic discrimination capability in nonurban environments. For the second image, GLMCA-Net produced complete and structurally clear building contours, while comparison models generally exhibited issues such as buildings merging into a single area and blurred boundaries. For the third image, our GLMCA-Net was the sole model that successfully identified the complete narrow path. This further validates GLMCA-Net's advantages in small object detection, particularly in terms of boundary clarity and maintaining continuity, demonstrating greater stability and reliability. This capability is crucial for segmenting both urban road networks and rural paths. These visualization results further validate GLMCA-Net's robustness in complex scenarios and its excellent perception of small objects and boundary details, fully demonstrating the effectiveness of its structural design.

### D. Ablation Study

To validate the contribution of each module to the overall performance, we conducted systematic ablation studies on the Vaihingen dataset, Potsdam dataset, and LoveDA dataset. Specifically, we started with the most basic module and gradually added key components: from MTB and ResB in the dual encoder to the three core blocks, DPCF block, AFE block, and SAEU block. Experimental results on three datasets are presented in Tables IV–VI.

Results on the Vaihingen dataset are shown in Table IV. Method A, which incorporates only MTB paths to capture long-range dependencies, exhibits overall lower performance with mIoU and m$F1$ of 68.89% and 80.26%, respectively. In particular, its IoU for the car category is only 39.94%, indicating that relying solely on global modeling is insufficient for capturing small-scale details. Method B, utilizing only ResB, significantly outperforms Method A, with mIoU improving to 80.41% and m$F1$ reaching 88.86%. The IoU for the car category notably surged to 70.54%, highlighting the advantage of the ResB in fine-grained detail extraction. Method C combines MTB and ResB to complement global and local information, achieving an mIoU of 82.04%, representing a substantial enhancement over single-path architectures. Building on this, the introduction of the DPCF block in Method D enhanced feature interaction through channel and spatial attention, elevating mIoU to 83.08% with steady improvements across all categories, validating its effectiveness in feature fusion. With the addition of the AFE block, Method E enhances modeling capabilities for multiscale and geometrically complex objects, achieving an mIoU of 84.29% and an m$F1$ of 91.28%. A notable gain of 2.75% IoU was observed for the car category, underscoring the AFE block's role in geometric modeling and weak feature enhancement. Method F, which integrates the SAEU block, achieves peak performance with an mIoU of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
ABLATION STUDY ON THE FUSION MECHANISM OF THE DPCF BLOCK ON THE ISPRS VAIHINGEN DATASET

| Fusion mechanism | Params (M) | Imp. surf. | Building | Low. veg. | Tree | Car | mIoU (%) | mF1 (%) |
|---|---|---|---|---|---|---|---|---|
| CAB | 43.39 | 92.83 | 92.37 | 72.18 | 82.05 | 77.74 | 83.43 | 90.75 |
| SAB | 43.35 | 92.86 | 92.27 | 71.99 | 81.81 | 78.73 | 83.53 | 90.82 |
| FEU | 115.58 | 93.64 | 93.67 | 73.03 | 82.74 | 83.41 | 85.30 | 91.87 |
| DPCF(CAB and SAB series connection) | 49.66 | 93.24 | 92.67 | 72.52 | 82.34 | 82.68 | 84.69 | 91.52 |
| DPCF-concat | 52.93 | 93.50 | 93.05 | 72.61 | 82.35 | 83.83 | 85.06 | 91.74 |
| DPCF(Ours) | 49.66 | 93.80 | 93.43 | 74.07 | 83.13 | 84.57 | 85.80 | 92.19 |

85.80%, confirming its effectiveness in high-fidelity reconstruction of fine details.

To validate the stability of the aforementioned performance improvement trend, the same experiments were replicated on the larger and more complex Potsdam dataset, with the results shown in Table V. Method A, utilizing only MTB, showed limited performance with an mIoU of just 73.98%. After employing ResB, Method B boosted the mIoU to 82.01% and the m$F1$ to 89.95%. Method C, which integrates MTB and ResB, further enhanced mIoU to 83.39%, demonstrating the complementary effects of global and local features. The subsequent introduction of the DPCF block (Method D) raised the mIoU to 84.27%, with marked improvements in the low vegetation and building categories, indicating that the fusion module enhances interaction between different feature levels. Method E further incorporates an AFE block, boosting mIoU to 85.61% and m$F1$ to 92.10%, which bolstered the model's representational capacity for complex geometries and multiscale objects. Finally, after integrating the SAEU block in Method F, mIoU and m$F1$ further improved to 86.37% and 92.54%, respectively, achieving optimal performance across all categories. Particularly in the building and tree categories, boundaries became more distinct, validating the SAEU block's contribution to enhancing structural fidelity.

To further evaluate the model's generalization ability in more diverse and challenging scenarios, such as urban-rural mixed environments, we also conducted ablation studies on the LoveDA dataset. Despite the dataset's more pronounced domain differences and class imbalance issues, our method still exhibited a clear pattern of performance growth, as shown in Table VI. Both Method A and B performed poorly, with mIoU of 44.45% and 50.75%, respectively. With the progressive introduction of the dual-path encoder and core modules, model performance steadily improved, culminating in the full model (Method F) achieving an mIoU of 53.91%. The performance on the 'barren' class was particularly noteworthy, improving from 18.64% by UNetFormer to 22.40%, which robustly demonstrates the effectiveness of the proposed network in tackling complex backgrounds and cross-domain variations.

Furthermore, we conducted an in-depth study on the fusion mechanism of the DPCF module, with experimental results shown in Table VII. Compared to using CAB or SAB mechanisms individually, serial connection, concat fusion, and other established approaches (such as FEU [57]), our proposed parallel fusion strategy significantly improved segmentation performance (achieving mIoU of 85.80%) while maintaining a low parameter count. This further validates the efficiency and necessity of the DPCF block.

### E. Efficiency Analysis

When designing semantic segmentation models for remote sensing images, balancing accuracy and efficiency is always a key issue. To comprehensively evaluate the proposed GLMCA-Net's capability of balancing computational efficiency and model performance, we conducted a comparative analysis of multiple mainstream methods on the ISPRS Vaihingen dataset. The analysis examined their parameters (Params), floating point operations (FLOPs), memory consumption (Memory), training time (Time cost), and performance metrics (mIoU and m$F1$), with the results presented in Table VIII.

GLMCA-Net achieves optimal segmentation performance while maintaining a relatively compact model size and reasonable computational overhead. Specifically, GLMCA-Net has 49.66 M parameters, which is lower than most transformer-based or multibranch structure methods, such as MANet (54.85 M), LANet (55.51 M), and DC-Swin (66.95 M), while being significantly smaller than A2FPN (66.25 M). In terms of FLOPs, GLMCA-Net achieves 181.93 G, which represents a favorable level of computational efficiency. It is lower than PSPNet (287.63 G) and DeeplabV3+ (277.35 G), and significantly outperforms A2FPN (339.91 G) and MANet (294.61 G), while achieving higher segmentation accuracy with 85.80% mIoU and 92.19% m$F1$.

In terms of operational efficiency, GLMCA-Net exhibits a reasonable memory consumption of 17.65 GB, lower than both A2FPN (21.19 GB) and PSPNet (19.65 GB). In addition, the training time is remarkably brief, at only 0.96 h, which is comparable to the 0.95 h required by UNetFormer, further underscoring its exceptional computational efficiency. Compared to lightweight networks with fewer parameters, such as BANet, GLMCA-Net presents a slight increase in both parameters and computational cost. However, it achieves significant performance improvements, with gains of 4.49% in mIoU and 2.76% in m$F1$, demonstrating its superior segmentation capability. This indicates that our model architecture is better suited for fully exploiting multiscale features and integrating global and local information while maintaining reasonable computational resource consumption. Consequently, it gains advantages in more complex remote sensing segmentation tasks.

Compared to recently proposed high-performance models such as UNetFormer and RS3Mamba, GLMCA-Net also demonstrates an excellent balance between accuracy and efficiency. UNetFormer has 43.23 M parameters and 174.21 G

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
COMPARATIVE EFFICIENCY AND ACCURACY ANALYSIS WITH SOTA METHODS ON THE ISPRS VAIHINGEN DATASET

| Method | Params (M) | FLOPs (G) | Memory (G) | Time cost (h) | mIoU (%) | mF1 (%) |
|---|---|---|---|---|---|---|
| PSPNet [13] | 53.45 | 287.63 | 19.65 | 1.31 | 80.64 | 89.03 |
| DeeplabV3+ [12] | 42.40 | 277.35 | 16.32 | 1.26 | 80.37 | 88.82 |
| BANet [54] | 12.73 | 85.43 | 7.53 | 0.41 | 81.31 | 89.43 |
| MANet [51] | 54.85 | 294.61 | 17.54 | 1.23 | 82.88 | 90.44 |
| DC-Swin [35] | 66.95 | 258.10 | 16.46 | 1.04 | 83.24 | 90.61 |
| UNetFormer [55] | 43.23 | 174.21 | 17.08 | 0.95 | 83.43 | 90.75 |
| LANet [52] | 55.51 | 209.13 | 19.37 | 1.20 | 82.28 | 90.04 |
| A2FPN [53] | 66.25 | 339.91 | 21.19 | 1.47 | 83.28 | 90.65 |
| RS3Mamba [56] | 43.32 | 157.89 | 18.71 | 1.17 | 83.50 | 90.79 |
| GLMCA-Net (Ours) | 49.66 | 181.93 | 17.65 | 0.96 | 85.80 | 92.19 |

FLOPs, which is slightly smaller than GLMCA-Net, but it is 2.37% lower in mIoU and 1.44% lower in m$F1$. RS3Mamba is another efficient lightweight Mamba architecture with lower parameters and FLOPs, but its performance falls short of GLMCA-Net, with a 2.30% difference in mIoU and a 1.40% difference in m$F1$. These comparison results clearly demonstrate that GLMCA-Net enhances model performance while controlling resource consumption, making it highly practical and promising for widespread adoption.

## VI. CONCLUSION

To tackle a series of challenges in the semantic segmentation of remote sensing images, this article proposes a novel global–local multilevel feature cascade network, GLMCA-Net. The network employs a dual-path encoding structure, integrating the powerful global modeling capabilities of MTB with the efficient local feature extraction capabilities of ResB. The DPCF block enables cross-path guidance and feature fusion, enhancing the consistency of global and local semantic expressions. The introduction of the AFE block effectively enhances sensitivity for local complex boundaries and small objects such as roads and water. Experimental results show that GLMCA-Net achieves significantly superior performance on three publicly available remote sensing semantic segmentation datasets. Ablation experiments further validate the contribution of each module to performance, and efficiency analysis also demonstrates that GLMCA-Net maintains high accuracy while having a low number of parameters and computational overhead. In the future, we will further explore more efficient linear mechanisms and dynamic fusion strategies to further reduce computational costs.

## REFERENCES


[1] F. Bastani, P. Wolters, R. Gupta, J. Ferdinando, and A. Kembhavi, "SatlasPretrain: A large-scale dataset for remote sensing image understanding," presented at the *IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2023/html/Bastani_SatlasPretrain_A_Large-Scale_Dataset_for_Remote_Sensing_Image_Understanding_ICCV_2023_paper.html

[2] A. Jamali, S. K. Roy, J. Li, and P. Ghamisi, "TransU-Net+: Rethinking attention gated TransU-Net for deforestation mapping," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 120, Jun. 2023, Art. no. 103332, doi: 10.1016/j.jag.2023.103332.

[3] W. Dong, Y. Sun, and J. Luo, "Fine mapping of key soil nutrient content using high resolution remote sensing image to support precision agriculture in Northwest China," in *Proc. 8th Int. Conf. Agro-Geoinformatics (Agro-Geoinformatics)*, Jul. 2019, pp. 1–5, doi: 10.1109/Agro-Geoinformatics.2019.8820504.

[4] H. Wang, X. Gong, B. Wang, C. Deng, and Q. Cao, "Urban development analysis using built-up area maps based on multiple high-resolution satellite data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102500, doi: 10.1016/j.jag.2021.102500.

[5] K. Zhao, J. Liu, Q. Wang, X. Wu, and J. Tu, "Road damage detection from post-disaster high-resolution remote sensing images based on TLD framework," *IEEE Access*, vol. 10, pp. 43552–43561, 2022, doi: 10.1109/ACCESS.2022.3169031.

[6] H. Bandarupally, H. R. Talusani, and T. Sridevi, "Detection of military targets from satellite images using deep convolutional neural networks," in *Proc. IEEE 5th Int. Conf. Comput. Commun. Autom. (ICCCA)*, Oct. 2020, pp. 531–535, doi: 10.1109/ICCCA49541.2020.9250864.

[7] S. Liu et al., "Rotated multi-scale interaction network for referring remote sensing image segmentation," presented at the *IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/html/Liu_Rotated_Multi-Scale_Interaction_Network_for_Referring_Remote_Sensing_Image_Segmentation_CVPR_2024_paper.html

[8] X. Ma et al., "LOGCAN+: Adaptive local–global class-aware network for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4404216, doi: 10.1109/TGRS.2025.3541871.

[9] J. Wang et al., "GRA: Detecting oriented objects through group-wise rotating and attention," in *Proc. ECCV*, A. Leonardis, E. Ricci, S. Roth, O. Russakovsky, T. Sattler, and G. Varol, Eds., 2024, pp. 298–315, doi: 10.1007/978-3-031-72643-9_18.

[10] C. Szegedy et al., "Going deeper with convolutions," presented at the *IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015. [Online]. Available: https://www.cv-foundation.org/openaccess/content_cvpr_2015/html/Szegedy_Going_Deeper_With_2015_CVPR_paper.html

[11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," presented at the *IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_Residual_Learning_CVPR_2016_paper.html

[12] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," presented at the Eur. Conf. Comput. Vis. (ECCV), May 2018. [Online]. Available: https://openaccess.thecvf.com/content_ECCV_2018/html/Liang-Chieh_Chen_Encoder-Decoder_with_Atrous_ECCV_2018_paper.html

[13] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," presented at the *IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2017/html/Zhao_Pyramid_Scene_Parsing_CVPR_2017_paper.html

[14] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[15] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002. Accessed: Nov. 19, 2024. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2021/html/Liu_Swin_Transformer_Hierarchical_Vision_Transformer_Using_Shifted_Windows_ICCV_2021_paper

[16] W. Wang et al., "PVT v2: Improved baselines with pyramid vision transformer," *Comput. Vis. Media*, vol. 8, no. 3, pp. 415–424, Sep. 2022, doi: 10.1007/s41095-022-0274-8.

[17] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surv.*, vol. 54, no. 10s, pp. 1–41, Jan. 2022, doi: 10.1145/3505244.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.

[18] H. Guo, B. Du, L. Zhang, and X. Su, "A coarse-to-fine boundary refinement network for building footprint extraction from remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 240–252, Jan. 2022, doi: 10.1016/j.isprsjprs.2021.11.005.

[19] C. OuYang and H. Li, "BuildNext-net: A network based on self-attention and equipped with an efficient decoder for extracting buildings from high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 1–19, 2025, doi: 10.1109/JSTARS.2025.3553907.

[20] Y. Zhao, G. Wang, C. Tang, C. Luo, W. Zeng, and Z.-J. Zha, "A battle of network structures: An empirical study of CNN, transformer, and MLP," 2021, *arXiv:2108.13002*.

[21] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612, doi: 10.1109/TGRS.2023.3314641.

[22] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[23] E. Zhu, Z. Chen, D. Wang, H. Shi, X. Liu, and L. Wang, "UNetMamba: An efficient UNet-like mamba for semantic segmentation of high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025, doi: 10.1109/LGRS.2024.3505193.

[24] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Remote sensing image change detection with mamba," 2024, *arXiv:2406.04207*.

[25] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024, doi: 10.1109/LGRS.2024.3407111.

[26] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440. [Online]. Available: http://openaccess.thecvf.com/content_cvpr_2015/html/Long_Fully_Convolutional_Networks_2015_CVPR_paper.html

[27] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-319-24574-4_28

[28] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6877–6886. [Online]. Available: http://openaccess.thecvf.com/content/CVPR2021/html/Zheng_Rethinking_Semantic_Segmentation_From_a_Sequence-to-Sequence_Perspective_With_Transformers_CVPR_2021_paper.html

[29] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7242–7252. Accessed: Nov. 19, 2024. [Online]. Available: http://openaccess.thecvf.com/content/ICCV2021/html/Strudel_Segmenter_Transformer_for_Semantic_Segmentation_ICCV_2021_paper.html

[30] D. Han, X. Pan, Y. Han, S. Song, and G. Huang, "FLatten transformer: Vision transformer using focused linear attention," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2023/html/Han_FLatten_Transformer_Vision_Transformer_using_Focused_Linear_Attention_ICCV_2023_paper.html

[31] N. Park and S. Kim, "How do vision transformers work?," 2022, *arXiv:2202.06709*.

[32] H. Fan et al., "Multiscale vision transformers," presented at the Proc. *IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2021/html/Fan_Multiscale_Vision_Transformers_ICCV_2021_paper.html

[33] W. Lu et al., "A CNN-transformer hybrid model based on CSWin transformer for UAV image object detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1211–1231, 2023, doi: 10.1109/JSTARS.2023.3234161.

[34] Y. Wu, L. Tang, and S. Yuan, "Semantic segmentation model of multi-source remote sensing images was used to extract winter wheat at tillering stage," *Sci. Rep.*, vol. 15, no. 1, p. 14751, Apr. 2025, doi: 10.1038/s41598-025-98449-3.

[35] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2022.3143368.

[36] Q. Zhu et al., "Samba: Semantic segmentation of remotely sensed images with state space model," *Heliyon*, vol. 10, no. 19, Oct. 2024, Art. no. e38495, doi: 10.1016/j.heliyon.2024.e38495.

[37] T. Dao and A. Gu, "Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality," 2024, *arXiv:2405.21060*.

[38] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, Dec. 2024, pp. 103031–103063.

[39] T. Li et al., "DAMamba: Vision state space model with dynamic adaptive scan," 2025, *arXiv:2502.12627*.

[40] Y. Shi, M. Dong, M. Li, and C. Xu, "VSSD: Vision mamba with non-causal state space duality," 2024, *arXiv:2407.18559*.

[41] J. Ruan, J. Li, and S. Xiang, "VM-UNet: Vision mamba UNet for medical image segmentation," 2024, *arXiv:2402.02491*.

[42] J. Liu et al., "Swin-UMamba: Mamba-based UNet with ImageNet-based pretraining," in *Proc. MICCAI*, M. G. Linguraru, Q. Dou, A. Feragen, S. Giannarou, B. Glocker, K. Lekadir, and J. A. Schnabel, Eds., 2024, pp. 615–625.

[43] Z. Wang, J.-Q. Zheng, Y. Zhang, G. Cui, and L. Li, "Mamba-UNet: UNet-like pure visual mamba for medical image segmentation," 2024, *arXiv:2402.05079*.

[44] S. Wu, J. Li, L. Qi, Z. Liu, and X. Gao, "Remote sensing imagery scene classification based on spiking neural network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 2795–2798, doi: 10.1109/IGARSS47720.2021.9554883.

[45] D. Duan, P. Liu, B. Hui, and F. Wen, "Brain-inspired online adaptation for remote sensing with spiking neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5605818, doi: 10.1109/TGRS.2025.3527039.

[46] L.-Y. Niu, Y. Wei, and Y. Liu, "Event-driven spiking neural network based on membrane potential modulation for remote sensing image classification," *Eng. Appl. Artif. Intell.*, vol. 123, Aug. 2023, Art. no. 106322, doi: 10.1016/j.engappai.2023.106322.

[47] S. Yang, B. Linares-Barranco, Y. Wu, and B. Chen, "Self-supervised high-order information bottleneck learning of spiking neural network for robust event-based optical flow estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 4, pp. 2280–2297, Apr. 2025, doi: 10.1109/TPAMI.2024.3510627.

[48] J. Guo et al., "CMT: Convolutional neural networks meet vision transformers," presented at the *IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2022/html/Guo_CMT_Convolutional_Neural_Networks_Meet_Vision_Transformers_CVPR_2022_paper.html

[49] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," presented at the *IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018. Accessed: Nov. 19, 2024. [Online]. Available: https://openaccess.thecvf.com/content_cvpr_2018/html/Hu_Squeeze-and-Excitation_Networks_CVPR_2018_paper.html

[50] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.

[51] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607713, doi: 10.1109/TGRS.2021.3093977.

[52] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021, doi: 10.1109/TGRS.2020.2994150.

[53] R. Li, L. Wang, C. Zhang, C. Duan, and S. Zheng, "A2-FPN for semantic segmentation of fine-resolution remotely sensed images," *Int. J. Remote Sens.*, vol. 43, no. 3, pp. 1131–1155, Feb. 2022, doi: 10.1080/01431161.2022.2030071.

[54] L. Wang, R. Li, D. Wang, C. Duan, T. Wang, and X. Meng, "Transformer meets convolution: A bilateral awareness network for semantic segmentation of very fine resolution urban scene images," *Remote Sens.*, vol. 13, no. 16, p. 3065, Aug. 2021, doi: 10.3390/rs13163065.

[55] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022, doi: 10.1016/j.isprsjprs.2022.06.008.

[56] X. Ma, X. Zhang, and M.-O. Pun, "RS3Mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024, doi: 10.1109/LGRS.2024.3414293.

[57] J. Yuan, A. Zhu, Q. Xu, K. Wattanachote, and Y. Gong, "CTIF-Net: A CNN-transformer iterative fusion network for salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 3795–3805, May 2024, doi: 10.1109/TCSVT.2023.3321190.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 13:31:46 UTC from IEEE Xplore. Restrictions apply.